Damaris Goumtsop

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Education

George Mason University — Fairfax, VA

Bachelor of Science in Computer Science — *Expected Graduation: May 2027*

- Courses: Data Structures & Algorithms, Numerical Analysis, Operating Systems, Databases, Linear Algebra
- Leadership & Organizations: CodePath Tech Fellow, ColorStack, National Society of Black Engineers (NSBE)

Technical Experience

United States Senate — Washington, DC

Software Engineer Intern — *June 2025 – August 2025*

- Engineered a distributed data reconciliation pipeline using Python and optimized server-side SQL validation logic, reducing transaction verification and data processing latency by 82% across 20,000+ asset records.
- Designed and implemented automated data governance workflows and API integrations, ensuring strict transactional consistency and zero-trust security architecture across 150+ distributed system nodes.
- Optimized relational database schemas and programmed automated constraint-checking microservices, improving data integrity from 73% to 98% and ensuring rigorous auditability for enterprise financial compliance.

United States Senate — Washington, DC

IT Intern — *June 2024 – August 2024*

- Architected and deployed relational database validation scripts (Python/SQL) to automate legacy flat-file spreadsheet workflows, reducing large-scale data verification cycles by 78%.
- Engineered automated ETL pipelines to parse, normalize, and index high-volume physical asset infrastructure data across distributed networks, enforcing strict data normalization and relational integrity.
- Designed relational schemas and programmatic audit trails to secure tracking mechanisms, eliminating manual oversight and ensuring 100% compliance with network data standards.

NASA L'SPACE Academy — Remote

Systems Engineer Intern — *Sept 2023 – Dec 2023*

- Managed multi-dimensional dataset trade-offs by executing mathematical trade-off analyses balancing payload, power, and thermal metrics using quantitative systems engineering frameworks.

Technical Projects

Soccer Predictor — *Python, FastAPI, LightGBM, DuckDB, Pandas, NumPy, SciPy*

- Architected a complete data engineering and ML pipeline to predict Premier League match outcomes, utilizing an analytical ETL ingestion framework to stream multi-season datasets into a local DuckDB instance.
- Dropped execution latency to 0.07 seconds per 380-match season by vectorizing a nested negative log-likelihood loop using NumPy array operations to eliminate a performance-killing pandas .iterrows() bottleneck.
- Developed a feature-stacked LightGBM classifier trained on a dynamic relational matrix of team form, Elo ratings, and Dixon-Coles parameters to serve predictive batch arrays via an asynchronous FastAPI REST API.

Drift – Distributed API Gateway — Go, Redis, Docker, Prometheus, SSE

- Accomplished sub-100ms routing latency under 1,000+ concurrent requests by engineering an HTTP reverse proxy from first principles using Go's low-level net/http/httputil and custom RESTful access layers.
- Eliminated multi-replica race conditions by developing an atomic Redis Lua token-bucket rate limiter, implementing a thread-safe, in-memory sync.Map fallback to enforce system security and API access controls.
- Managed data streaming pipelines by deploying a Prometheus metrics engine and a low-overhead Server-Sent Events (SSE) data hub to stream real-time throughput analytics directly to an embedded dashboard.

Technical Skills

Languages: Python, Go, Java, SQL, C, C++, JavaScript, Lua
Frameworks & Libraries: FastAPI, React, Spring Boot, Node.js, LightGBM, Pandas, NumPy, SciPy
Databases: PostgreSQL, MySQL, Redis, DuckDB, SQLite
Infrastructure & Tools: Git, Docker, AWS, Linux, CI/CD, RESTful APIs, k6, Pytest, JUnit